Microprocessor

Chapter 7: Advanced Microprocessor

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- 80286:
  - ✓ Architecture (Block Diagram), Registers, (Real/Protected mode), Privilege Levels, Descriptor Cache, Memory Access in GDT and LDT, Multitasking, Addressing Modes, Flag Register
- 80386:
  - ✓ Architecture (Block Diagram), Register organization, Memory Access in Protected Mode, Paging (Up to LA to PA)

80286 Microprocessor

- The 80286 was designed for multi-user systems with multitasking applications, including communications and real-time process control.
- It has 134,000 transistors and consisted of four independent units: address unit, bus unit, instruction unit and execution unit.
- These were organized into a pipeline, significantly increasing performance.
- It was produced in a 68-pin package include PLCC(Plastic Leaded Chip Carrier),LCC(Leadless chip carrier) and PGA(Pin Grid Array) package.
- The Intel 80286 had a 24-bit address bus and was able to address up to 16MB of RAM, compared to 1MB for its predecessor.
- However cost and initial rarity of software using the memory above 1 MB meant that 80286 computers were rarely shipped with more than one megabyte of RAM.

80286 Microprocessor

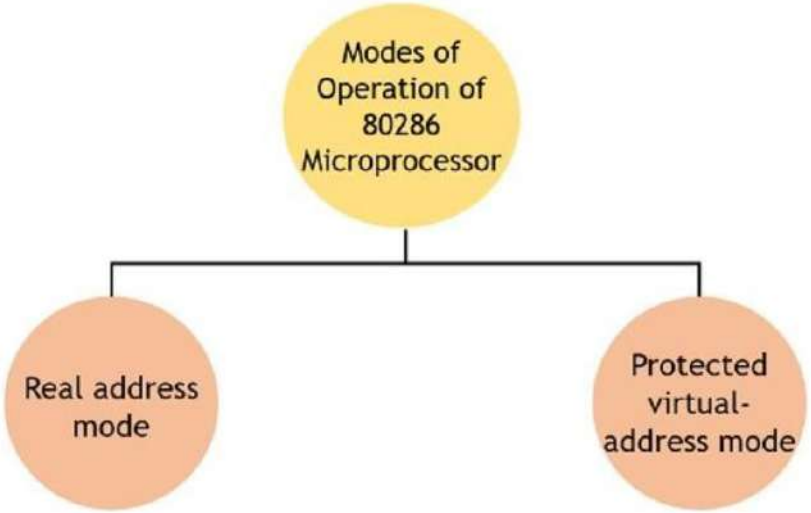
- The 80286 has built-in memory protection that supports operating system and task isolation as well as program and data privacy.
- A 12 MHz 80286 provides about six times more than the 5 MHz 8086.
- The 80286 includes memory management capabilities that map 2^{30} (one gigabyte) of virtual address space per task into 2^{24} bytes (16 megabytes) of physical memory.

Operating modes of 80286 microprocessor

- **Intel 80286 has 2 operating modes.**
  - Real Address Mode
    - ✓ 80286 just act as a faster version of 8086
    - ✓ Program for 8086 can be executed without modification in 80286
    - ✓ It offers memory addressability of 1 MB of physical memory.
  - Protected Virtual Address Mode
    - ✓ Supports multitasking because multiple programs can be executed using virtual memory.
    - ✓ Able to run several program at the same time
    - ✓ This mode of 80286 offers memory addressability of 16 MB of physical memory along with 1 GB of virtual memory

Operating modes of 80286 microprocessor

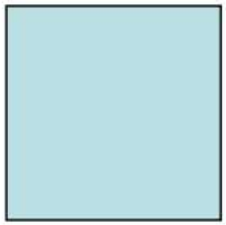
- As using virtual memory, space for other programs can be saved. Sometimes bulky programs also do exist that cannot be stored in physical memory, so virtual memory is utilized in order to execute large programs.
- This mode is used in 80286, so that in case of memory failure in real address mode, it can stay in protected manner.



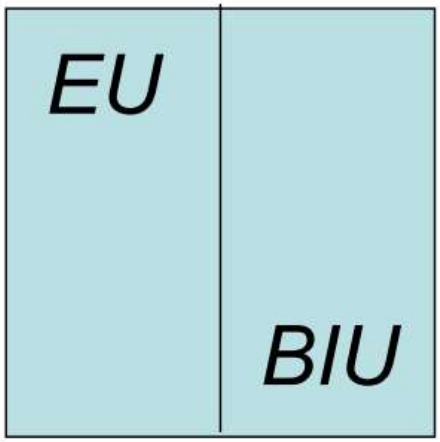
Basic Features

- **More Addressable Memory**
  - ✓ In protected mode 80286 can address 16 megabytes of physical memory
  - ✓ Whereas 8086 can address only 1 megabyte
- **Virtual Memory in Protected Mode**
  - ✓ 80286 can treat external storage as it were physical memory
  - ✓ Execute programs that are too large to be contained in physical memory
  - ✓ Program can be up to 2^{30} Bytes
- **Four Separate Processing/Functional Parts**
  - ✓ Address unit
  - ✓ Bus unit
  - ✓ Instruction unit
  - ✓ Execution unit

Internal Block Diagram of 80286



8085



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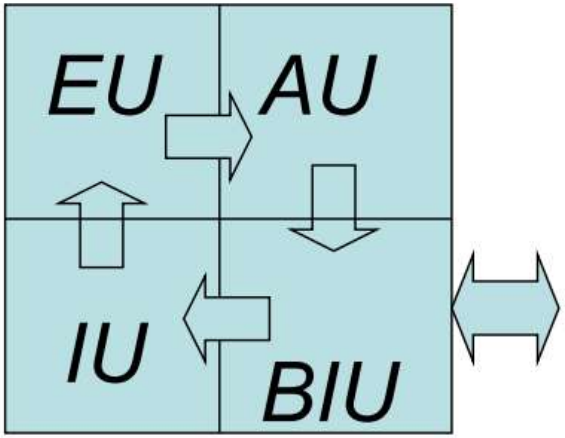
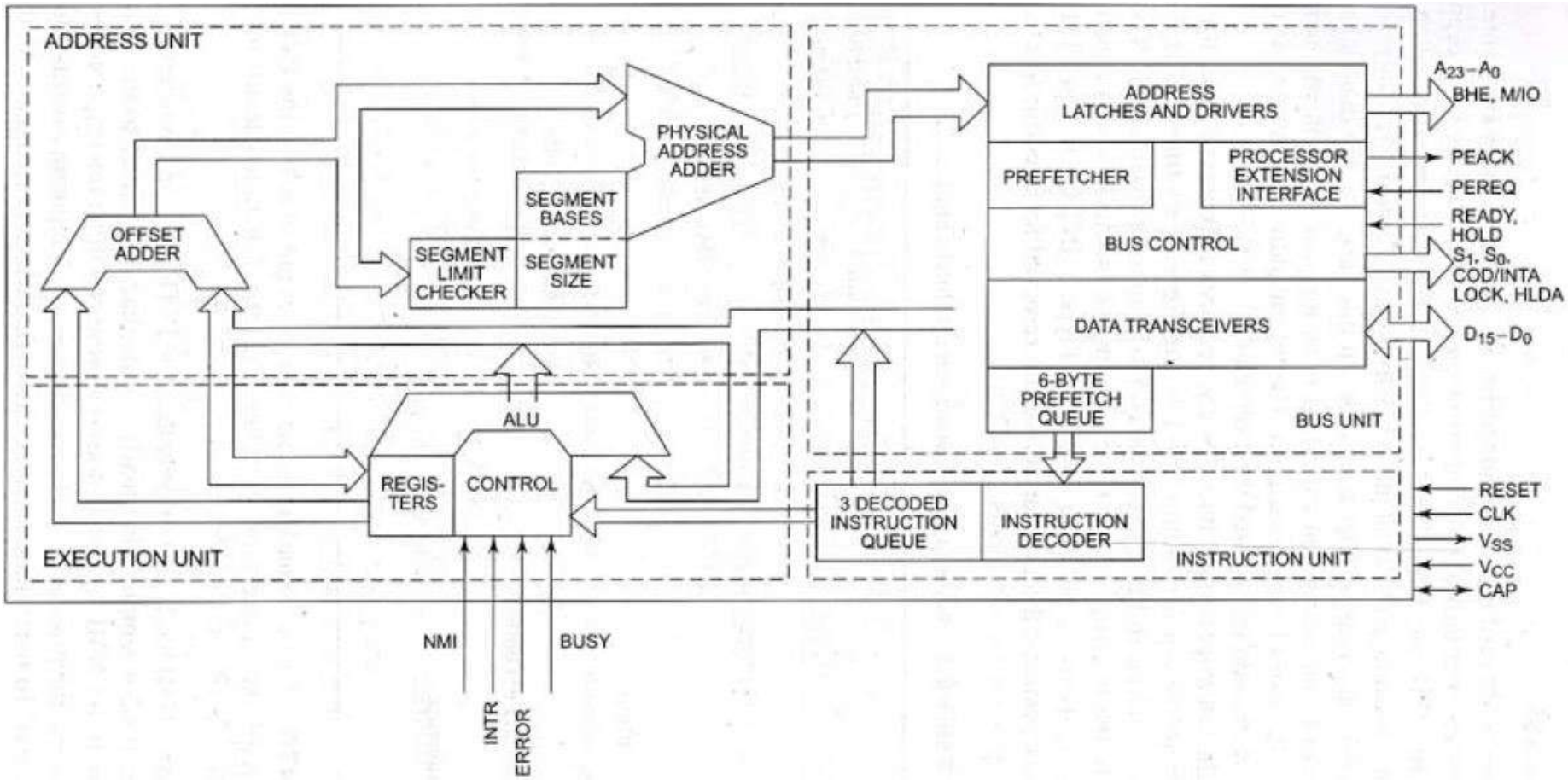
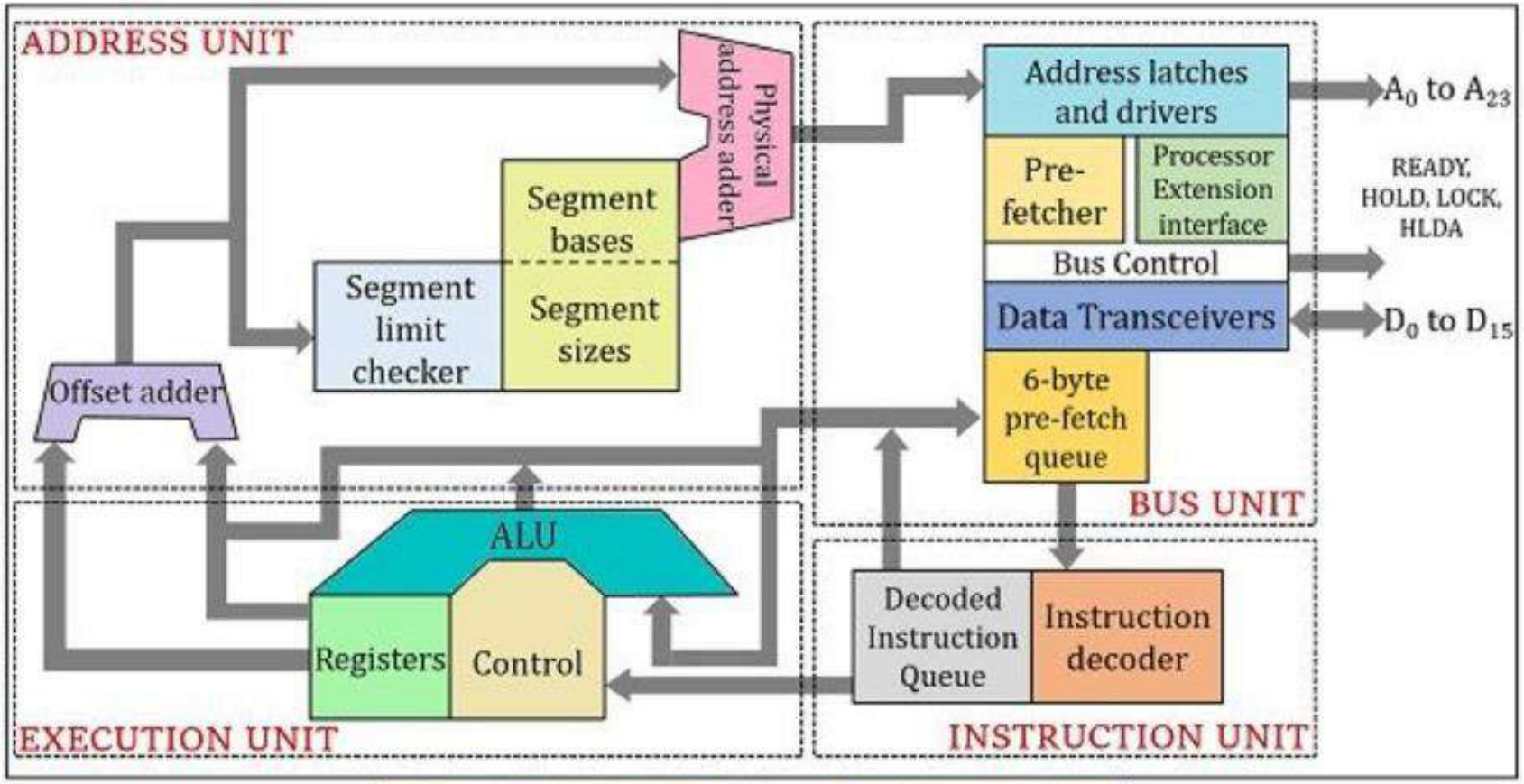


80286

80286 Microprocessor Architecture



80286 Microprocessor Architecture



Block Diagram of 80286 Microprocessor

80286 Microprocessor Architecture

Bus Unit (BU):

It has address latches, data transceivers, bus interface and circuitry, instruction pre-fetcher, processor extension interface and 6 byte instruction queue.

Functions :

- To perform all memory and I/O read and write.
- To pre-fetch the instruction bytes.
- To control the transfer of data to and from processor extension devices like 80287 math co-processor.
- Whenever BU is not using the buses for the operation, it pre-fetches the instruction bytes and put them is a 6 byte pre-fetch queue.

80286 Microprocessor Architecture : BUS Unit

- Performs all memory and I/O read and write operations.
- Take care of communication between CPU and a coprocessor.
- Transmit the physical address over address bus A0 – A23.
- Prefetcher module in the bus unit performs this task of prefetching.
- Bus controller controls the prefetcher module.
- Fetched instructions are arranged in a 6-byte prefetch queue.

80286 Microprocessor Architecture : Execution Unit

- EU executes the instructions received from the decoded instruction queue sequentially.
- Contains Register Bank.
- contains one additional special register called Machine status word (MSW) register - lower 4 bits are only used.
- ALU is the heart of execution unit.
- After execution ALU sends the result either over data bus or back to the register bank.

80286 Microprocessor Architecture

Execution Unit (EU):

It includes ALU, registers and the Control unit. Registers are general purpose, index, pointer, flag register and 16 –bit Machine Status Word (MSW).

Functions :

- To sequentially execute the instructions received from the instruction unit.
- ALU result is either stored in register bank or sent over the data bus.

80286 Microprocessor Architecture : Instruction Unit

- Receive arranged instructions from 6 byte prefetch queue.
- Instruction decoder decodes up to 3 prefetched instruction and are latched them onto a decoded instruction queue.
- Output of the decoding circuit drives a control circuit in the Execution unit.

80286 Microprocessor Architecture

Instruction Unit (IU):

It has 3 decoded instruction queue and instruction decoder.

Functions :

- It fully decodes up to three prefetched instructions and holds them in a queue.
- So that EU can access them.
- It helps the processor to speed up, as pipelining of instruction is done.

80286 Microprocessor Architecture : Address Unit

- Calculate the physical addresses of the instruction and data that the CPU want to access
- Address lines derived by this unit may be used to address different peripherals.
- Physical address computed by the address unit is handed over to the BUS unit.

80286 Microprocessor Architecture

Address Unit (AU):

It consists of segment registers, offset address and a physical address adder.

Functions :

- Compute the physical address that will be sent out to the memory or I/O by BU.
- 80286 operate in two different modes

  1. real address mode

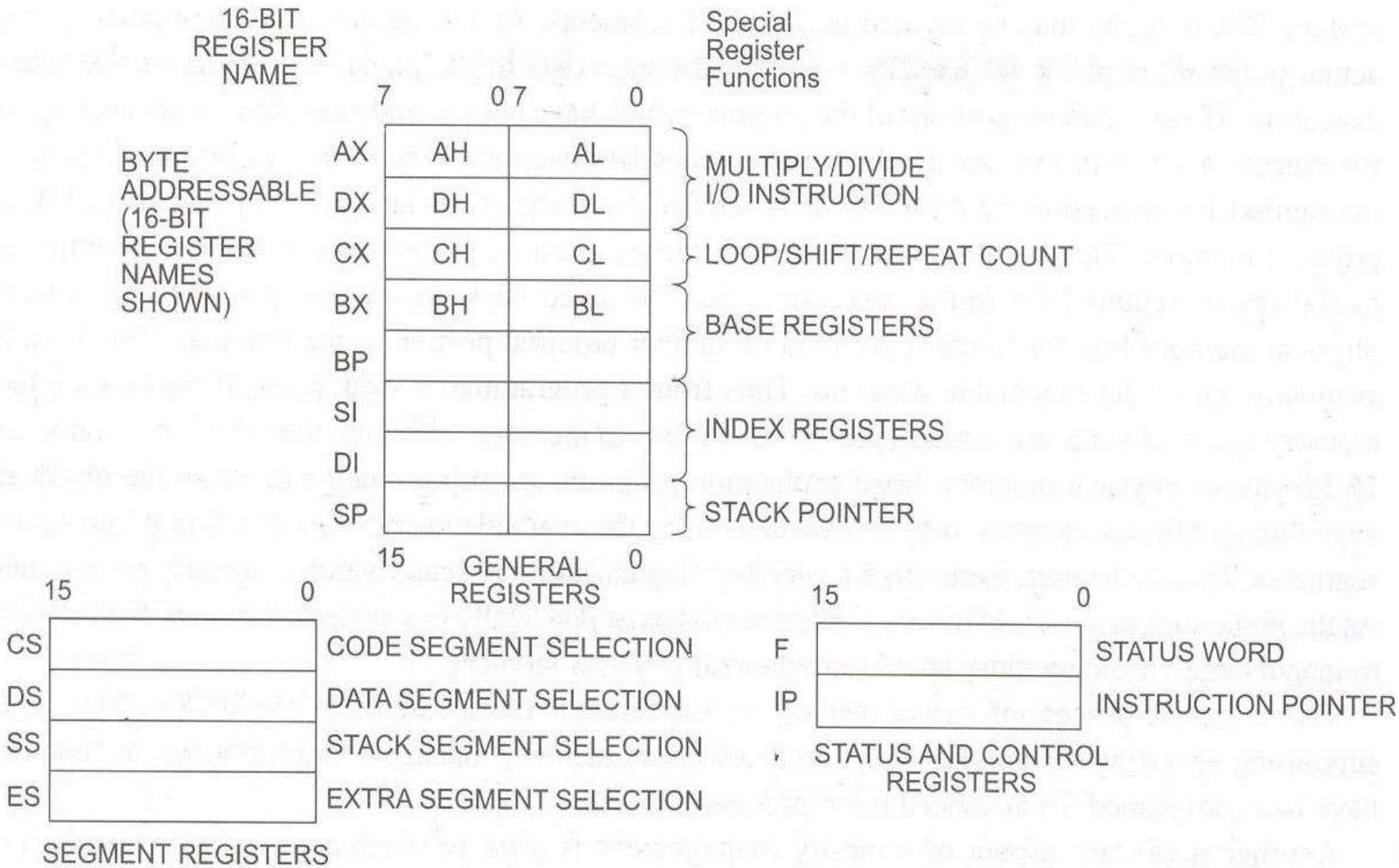
  2. Protected virtual address mode.
- When used in Real address mode, AU computes the address with segment base and offset like 8086. Segment register are CS, DS, ES and SS hold base address. IP, BP, SI, DI , SP hold offset.
- Maximum physical space allowed in this mode is 1MB.
- When 80286 operate in protected mode, the address unit acts as MMU.
- All 24 address lines used and can access up to 16MB of physical memory.
- If descriptor table scheme is used it can address up to 1GB of virtual memory.

Register organization of 80286

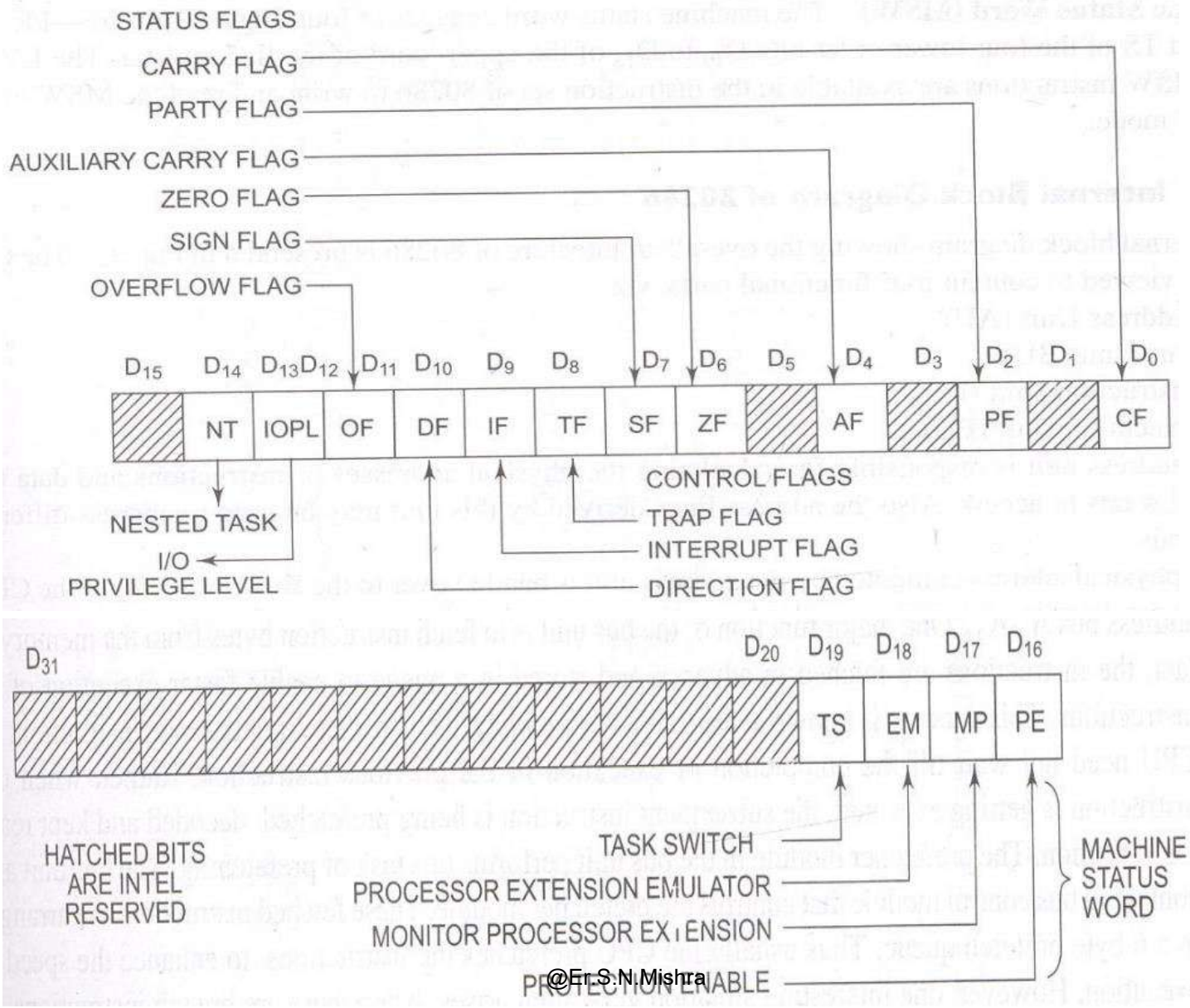
- **The 80286 CPU contains almost the same set of registers, as in 8086:**

1. Eight 16-bit general purpose registers.
2. Four 16 bit segment registers.
3. Status and control register.
4. Instruction pointer.

Register Organization of 80286



Flag Registers in 80286



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Flags in 80286

- 80286 Processor consist of 32 bit flag register the lower order 16 bits(D0 to D15) represents different flags and higher order 16 bits(D16 to D31) represents machine Status Word.
- There are 11 flags in 80286 processor:
- **C (Carry)**
  - ✓ The carry flag shows that the arithmetic or logic operation generated a carry/borrow.
  - ✓ If C=1, there is carry; if C= 0, there is no carry.
- **P (parity)**
  - ✓ Parity is logic 0 for odd parity and logic 1 for even parity. Parity is a count of ones in a number expressed as even or odd.
- **A (auxiliary carry)**
  - ✓ The auxiliary carry holds the carry (half-carry) after addition or borrow after subtraction between bits positions 3 and 4 of the result. This highly specialized flag bit is tested by the DAA and DAS instructions to adjust the value of AL after a BCD addition or subtraction.
- **Z (zero)**
  - ✓ The zero flag shows that the result of an arithmetic or logic operation is zero. If Z=1, the result is zero; if Z= 0, the result is not zero.

Flags in 80286

- **S (sign)**
  - ✓ The sign flag holds the arithmetic sign of the result after arithmetic or logic instruction executes. If S=1, the sign bit (leftmost hit of a number) is set or negative; if S=0, the sign bit is cleared or positive.
- **T (trap)**
  - ✓ The trap flag enables trapping through an on-chip debugging feature. (A program is debugged to find an error or bug.) If the T flag is enabled (1), the microprocessor interrupts the flow of the program on conditions as indicated by the debug registers and control registers. If the T flag is at logic 0, the trapping (debugging) feature is disabled.
- **I (interrupt)**
  - ✓ The interrupt flag controls the operation of the INTR (interrupt request) input pin. If I=1,the INTR pin is enabled: if I= 0, the INTR pin is disabled.
  - ✓ The state of I flag bit is controlled by the STI (set I flag) and CLI (clear I flag) instructions.
- **D (direction)**
  - ✓ The direction flag selects either the increment or decrement mode for the DI and/or SI registers during string instructions.
    - ✓ If D=1, the registers are automatically decremented
    - ✓ If D=0, the registers are automatically incremented. The D flag is set with the STD (set direction) and cleared with the CLD (clear direction) instructions.

Flags in 80286

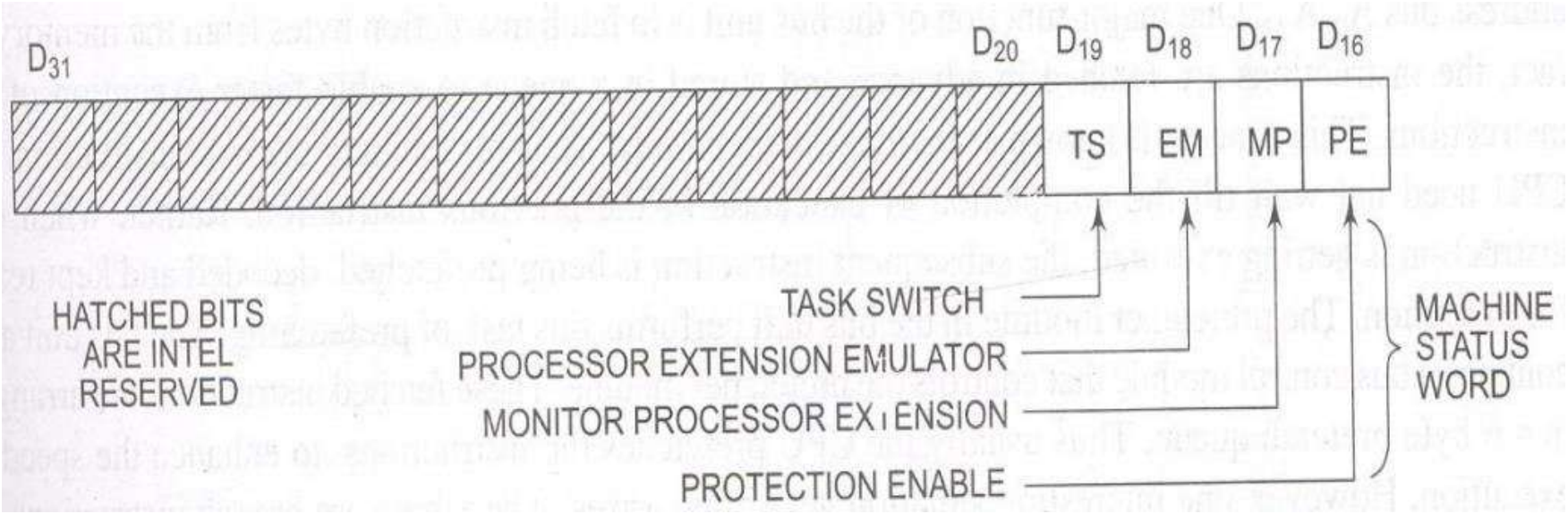
- **0 (Overflow)**
  - ✓ Overflows occur when signed numbers are added or subtracted. An overflow indicates that the result has exceeded the capacity of the machine.
  - ✓ For unsigned operations, the overflow flag is ignored.
- **IOPL (I/0 privilege level Bit D12 and D13)**
  - ✓ IOPL is used in protected mode operation to select the privilege level for I/O devices.
  - ✓ If the current privilege level is higher or more trusted than the IOPL, I/O executes without hindrance.
  - ✓ If the IOPL is lower than the current privilege level, an interrupt occurs, causing execution to suspend.
  - ✓ Note that an IOPL of 00 is the highest or most trusted: if IOPL is 11, it is the lowest or least trusted.
- **NT (Nested task)**
  - ✓ The nested task flag indicates that the current task is nested within another task in protected mode operation.
  - ✓ This line is set when the task is nested by software.
  - ✓ When set, it indicates that one system task has invoked another through a CALL instruction as opposed to a JMP.
  - ✓ For multitasking this can be manipulated to our advantage

Machine Status Word

- *The Machine Status indicates the protection mode of the processor and the presence and status of numeric coprocessor.*
- *It is set to Zero (Real Mode) by the reset signal, while setting PE (Protection Enabled) to one indicates entry to protected virtual mode.*

Machine Status Word

- Consist of four flags.
- These are – PE, MP, EM and TS
- LMSW & SMSW instruction are available in the instruction set of 80286 to write and read the MSW in real address mode.



Machine Status Word

- PE -Protection enable
- ✓ Protection enable flag places the 80286 in protected mode, if set.
- ✓ This can only be cleared by resetting the CPU.
- MP-Monitor processor extension
- ✓ Flag allows WAIT instruction to generate a processor extension.
- EM – Emulate processor extension flag
- ✓ If set , causes a processor extension absent exception and permits the emulation of processor extension by CPU.
- TS – Task switch
- ✓ If set this flag indicates the next instruction using extension will generate exception 7, permitting the CPU to test whether the current processor extension is for current task.

Operating Modes of 80286: Real Address Mode

- Act as a fast 8086
- Instruction set is upwardly compatible
- It address only 1 M byte of physical memory using A0-A19.
- In real addressing mode of operation of 80286, it just acts as a fast 8086.
- The instruction set is upward compatible with that of 8086.
- The 80286 addresses only 1Mbytes of physical memory using A0- A19.
- The lines A20-A23 are not used by the internal circuit of 80286 in this mode.
- In real address mode, while addressing the physical memory, the 80286 uses BHE along with A0- A19.
- The 20-bit physical address is again formed in the same way as that in 8086.

Operating Modes of 80286: Real Address Mode

- The contents of segment registers are used as segment base addresses.
- The other registers, depending upon the addressing mode, contain the offset addresses. Because of extra pipelining and other circuit level improvements, in real address mode also, the 80286 operates at a much faster rate than 8086, although functionally they work in an identical fashion.
- As in 8086, the physical memory is organized in terms of segments of 64Kbyte maximum size.
- An exception is generated, if the segment size limit is exceeded by the instruction or the data.
- The overlapping of physical memory segments is allowed to minimize the memory requirements for a task.
- The 80286 reserves two fixed areas of physical memory for system initialization and interrupt vector table.

Operating Modes of 80286: Real Address Mode

- In the real mode the first 1Kbyte of memory starting from address 0000H to 003FFH is reserved for interrupt vector table.
- Also the addresses from FFFF0H to FFFFFH are reserved for system initialization.
- The program execution starts from FFFFH after reset and initialization.
- The interrupt vector table of 80286 is organized in the same way as that of 8086.
- Some of the interrupt types are reserved for exceptions, single-stepping and processor extension segment overrun, etc.
- When the 80286 is reset, it always starts the execution in real address mode.
- In real address mode, it performs the following functions: it initializes the IP and other registers of 80286, it prepares for entering the protected virtual address mode.

Operating Modes of 80286: Real Address Mode

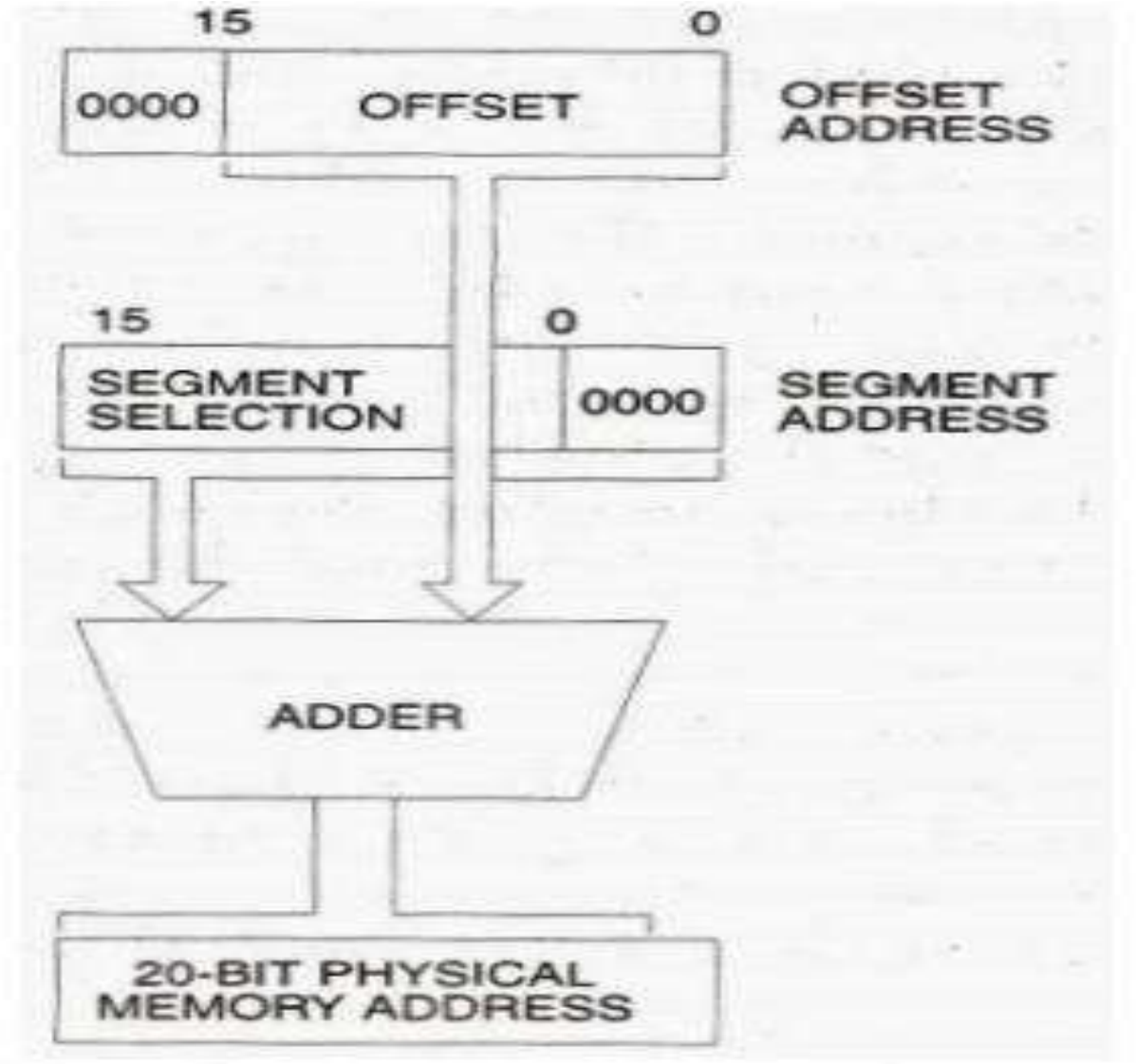


Figure: Real Address Mode Address Calculation

Operating Modes of 80286: Protected Virtual Address Mode

- 80286 is the first processor to support the concepts of virtual memory and memory management.
- The virtual memory does not exist physically it still appears to be available within the system.
- The concept of VM is implemented using Physical memory that the CPU can directly access and secondary memory that is used as a storage for data and program, which are stored in secondary memory initially.
- The 80286 is able to address 1 G byte (230 bytes) of virtual memory per task.
- The complete virtual memory is mapped on to the 16Mbyte physical memory.
- If a program larger than 16Mbyte is stored on the hard disk and is to be executed, if it is fetched in terms of data or program segments of less than 16Mbyte in size into the program memory by swapping sequentially as per sequence of execution.
- 80286 uses the 16-bit content of a segment register as a selector to address a descriptor stored in the physical memory.

Operating Modes of 80286: Protected Virtual Address Mode

- The descriptor is a block of contiguous memory locations containing information of a segment, like segment base address, segment limit, segment type, privilege level, segment availability in physical memory, descriptor type and segment use another task.
- Hardware reset is the only way to come out of protected mode.

Operating Modes of 80286: Protected Virtual Address Mode

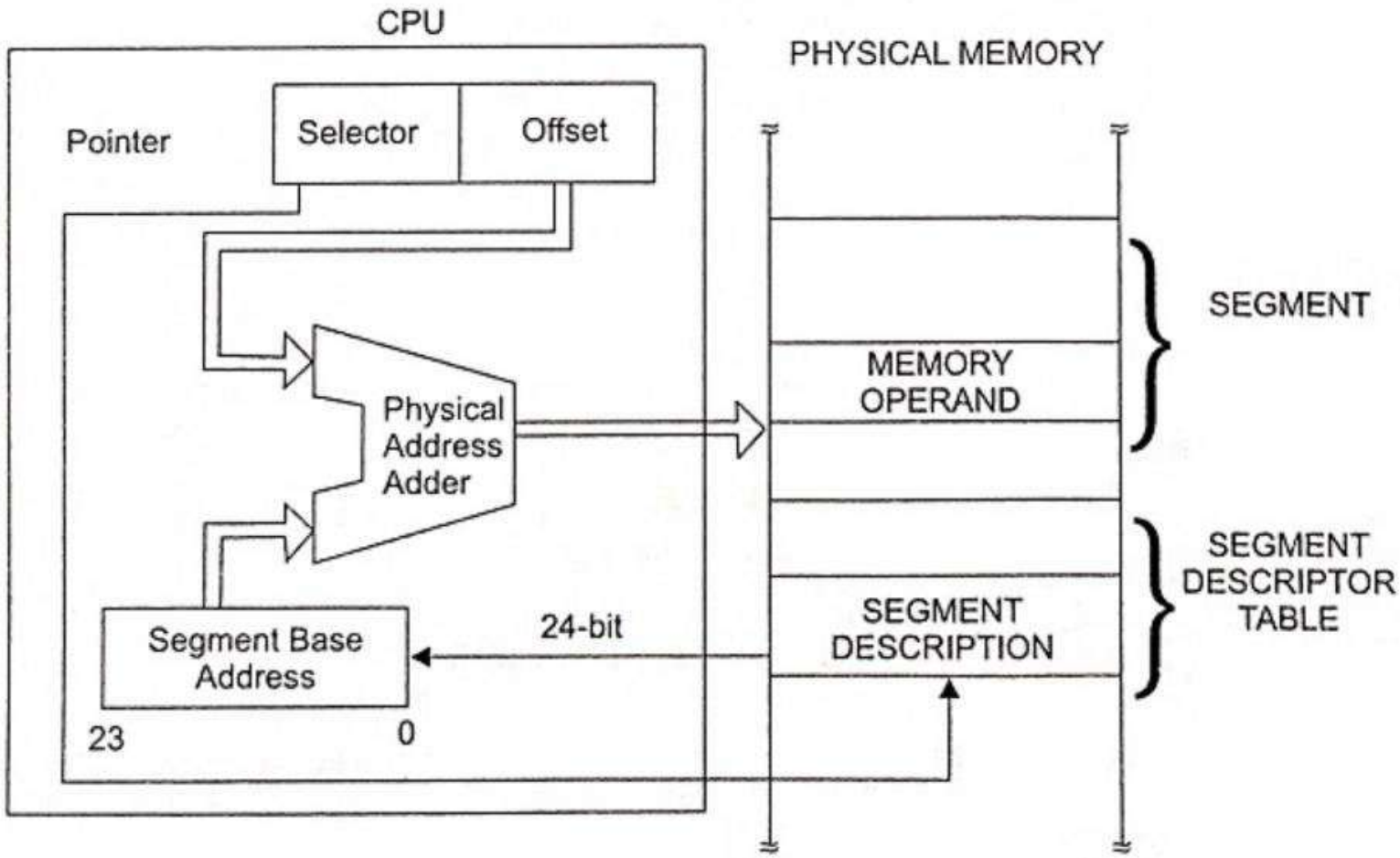


Fig: Physical Address Calculation in PVAM

80286 Privilege Levels

- **There are four Privilege Levels:**
- **00: Kernel Level (Highest Privilege Level)**
- **01: OS Services**
- **10: OS Extensions**
- **11: Applications (Lowest Privilege Level)**

80286 Privilege Levels

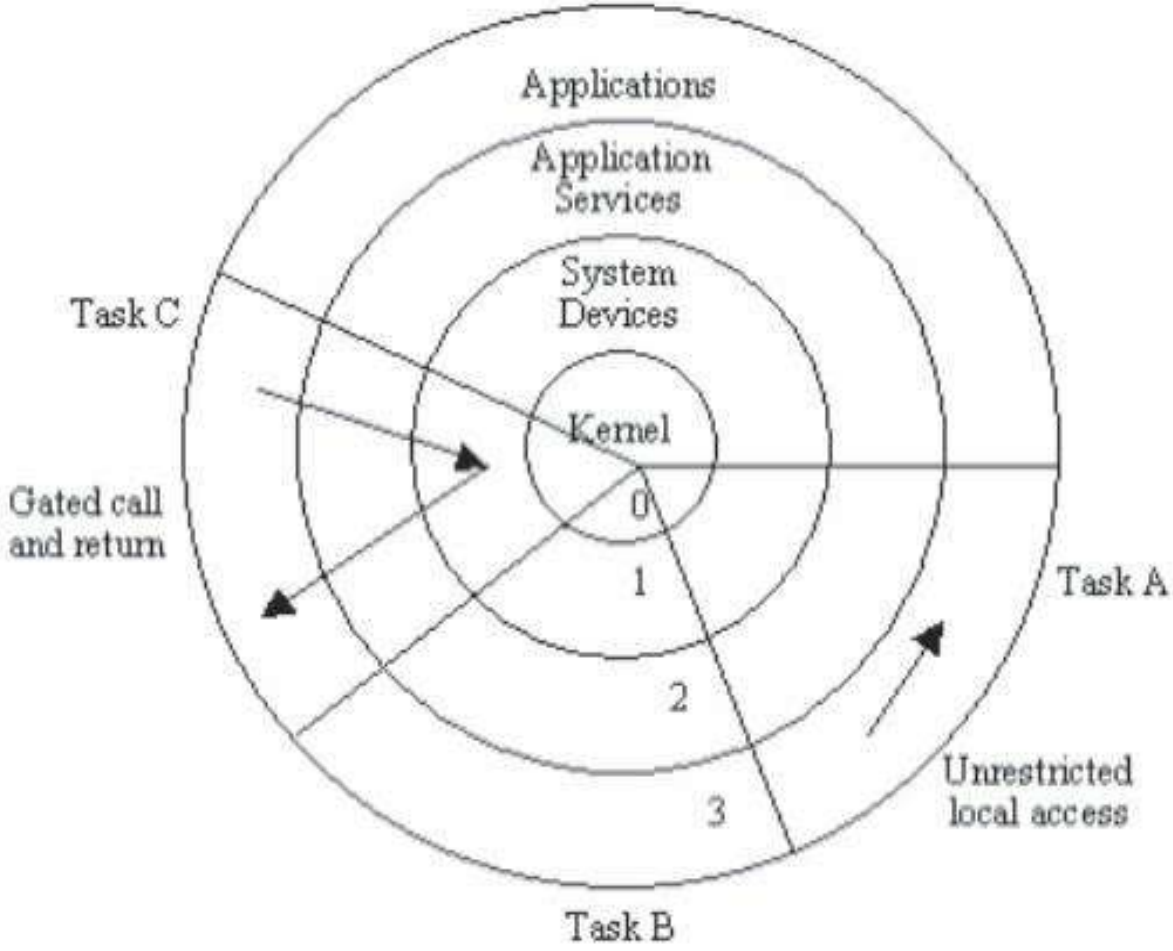


Figure: Privilege Level

80286 Privilege Levels

Each task assigned a privilege level, which indicates the priority or privilege of that task.

It can only changed by transferring the control, using gate descriptors, to a new segment.

A task executing at level 0, the most privileged level, can access all the data segment defined in GDT and LDT of the task.

A task executing at level 3, the least privileged level, will have the most limited access to data and other descriptors.

The use of rings allows for system software to restrict tasks from accessing data.

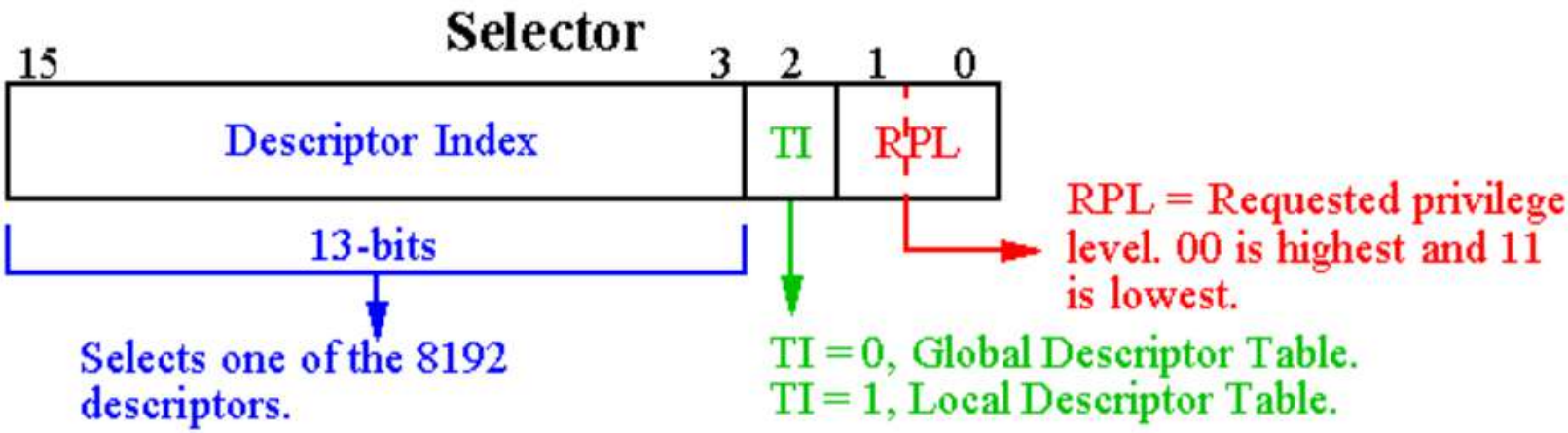
In most environments, the operating system and some device drivers run in ring 0 and applications run in ring 3.

Virtual Address, Descriptor and Descriptor Table

- In the 80286 protected modes, programs use only virtual addresses, without any access to the real addresses generated by the translation unit within the CPU.
- The virtual address is always composed of two parts, the displacement and the segment selector.
- The displacement is a 16-bit value, and is sometimes called an offset (within a segment) or an effective address.
- The segment selector specifies exactly which segment contains the data to be accessed.
- The selector is also a 16-bit item made up of three distinct fields which are described later.
- The (displacement, selector) pair together makes a 32-bit pointer, which represents the full virtual address of the object to be accessed.

Virtual Address, Descriptor and Descriptor Table

- In protected mode, the selector is made up of three fields, shown in figure:



Virtual Address, Descriptor and Descriptor Table

- In this format, only the 14 most significant bits are used in the address calculation.
- We can thus have maximum of 2^{14} or 16384 different segments, which since each bytes gives us a virtual memory space of 2^{30} bytes, or 1 gigabyte. The least significant of these 14 bits is the Table Indicator (TI) bit, which selects between two segment descriptor tables. Each table may have up to 2^{14} entries, describing up to 8192 segments. One of these tables, the Global Descriptor Table (GOT), is shared between all of the programs running in the machine.
- The second, the Local Descriptor Table (LOT), is local to the currently executing task or context, so that the virtual address space of 1 gigabyte is a per-task limit. Since each task may have its own LOT, it may have a "private" address space of up to 1 gigabyte and a "global" address space, shared with all other tasks currently executing, of another 1gigabyte.
- The global space is commonly used for the operating system, for shared library routines and data, and for access to global resources of the system. Providing one local segment descriptor table per task enables the separation of data belonging to different tasks, giving an increase in the reliability of the system, and reducing the consequences of programming errors.

Virtual Address, Descriptor and Descriptor Table

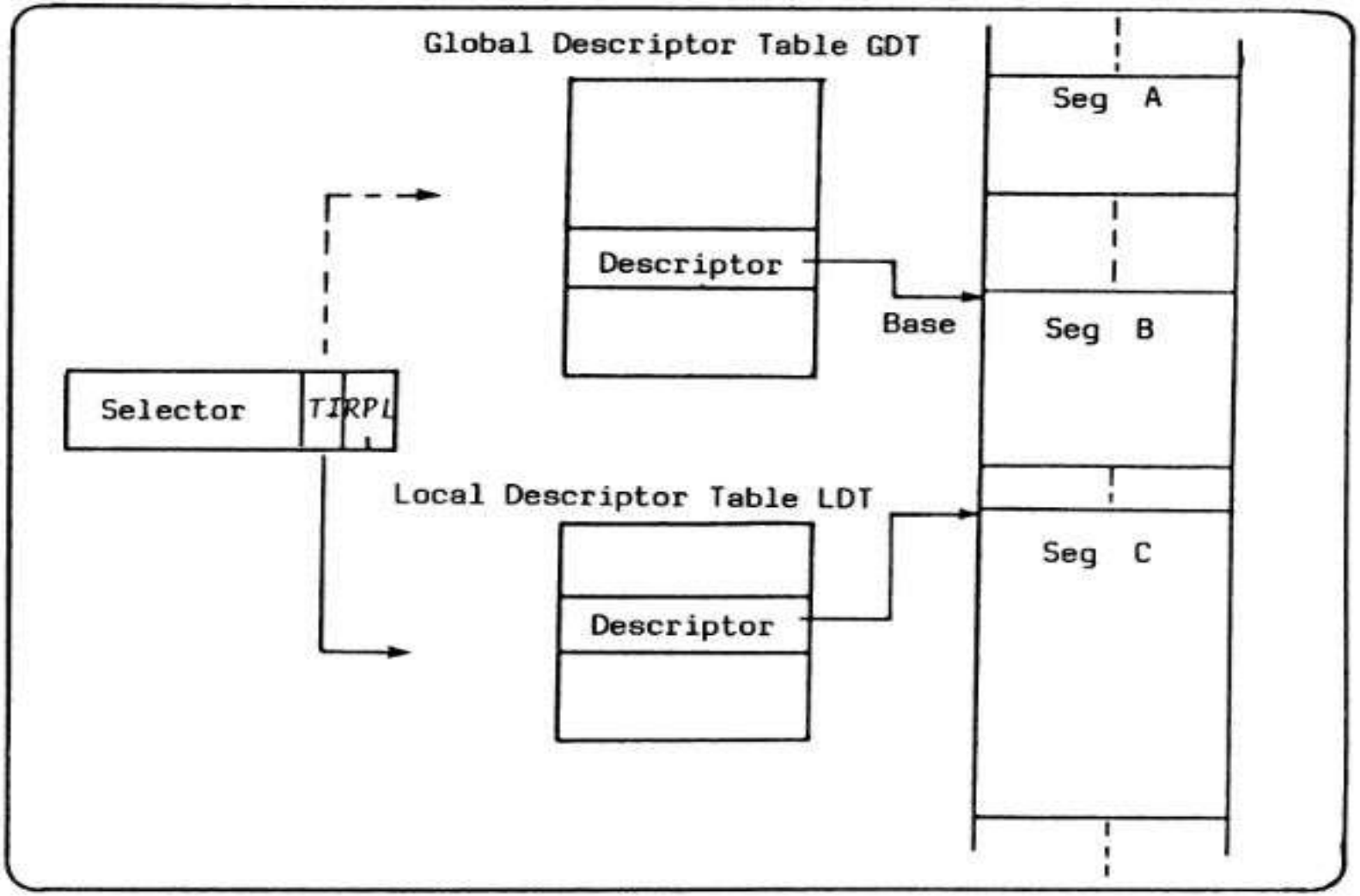


Figure 3.1.3 Descriptor table selection by the TI bit

Descriptors and Descriptor Table

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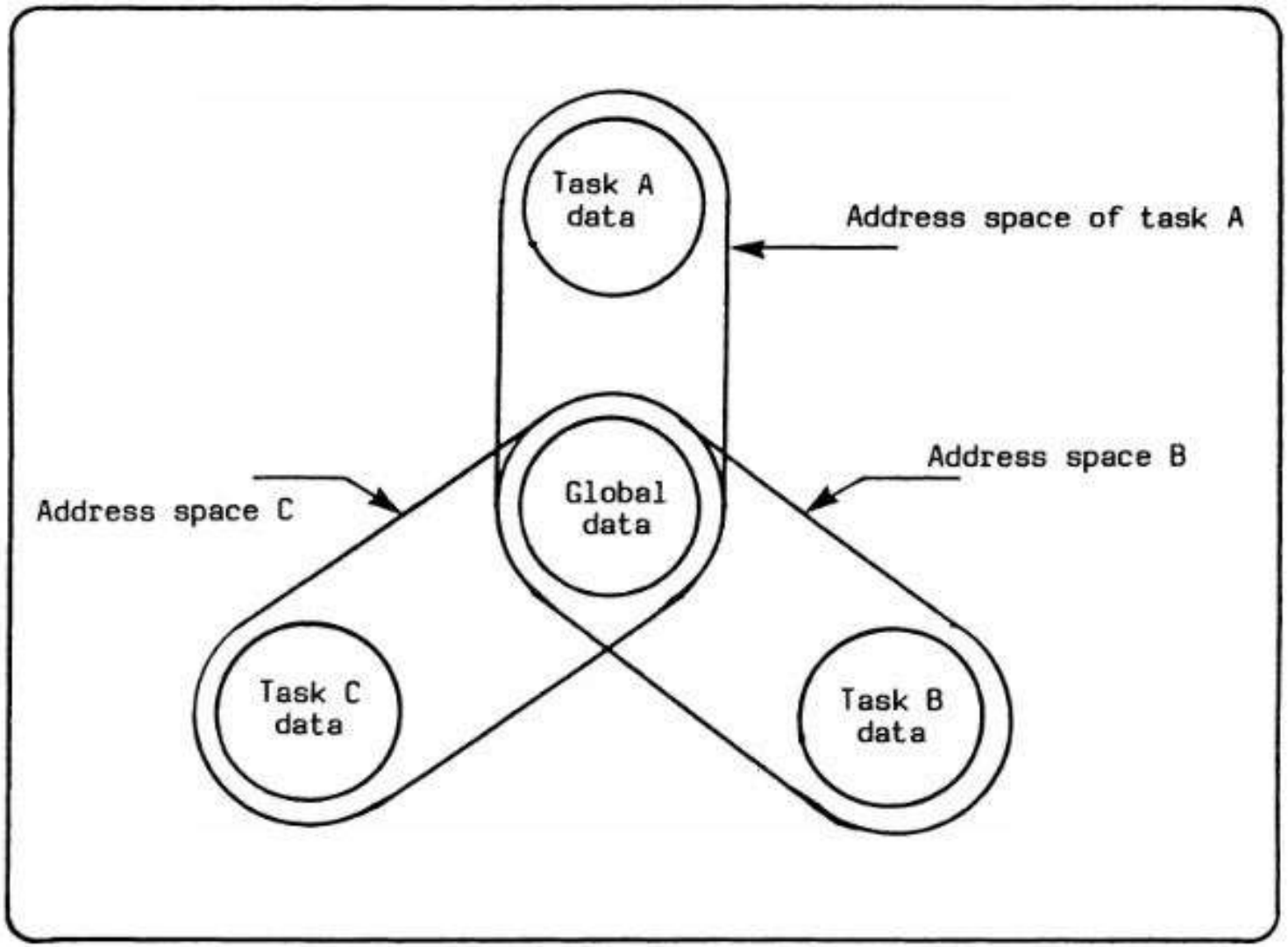
Virtual Address, Descriptor and Descriptor Table

- A table of descriptors, indexed by the segment selector, must be resident in main memory during the task's execution.
- The table is used as an indirection table for the management of protection and for the generation of physical addresses for every memory access.
- It serves as the means of translating a 32-bit virtual pointer into a 24-bit physical address.
- In its execution, each task may reference the GDT and the LDT.
- Nevertheless, it is possible to envisage an application which only uses the GDT.
- This might simplify understanding the addressing mechanisms that are in use, but it would also destroy the principle of separating the data between tasks, and thus diminish the protection between programs.
- In this type of system, it would be impossible to limit or forbid task A from accessing the data which logically belongs to an independent task B.
- In the 80286 system, maximum flexibility is provided by delaying the construction of the descriptor tables until after all the stages of compilation, and using

Virtual Address, Descriptor and Descriptor Table



Virtual Address, Descriptor and Descriptor Table

- A segment descriptor table consists of a vector of 8-byte quantities, called descriptors. A table may have up to 8192 entries, so a full-sized table exactly fills one segment.
- The 80286 system provides two different types of descriptors, segment descriptors, and system descriptors, used for different control operations, such as task switching, interrupt entries, or for describing special types of data segments.
- The largest field of a descriptor is the 24-bit base address, which is the real address of the start of the segment in main memory. A second field gives the real physical size of the segment, which must be less than or equal to 64 kilobytes.
- The remaining byte contains a number of interesting fields, used in the management of memory and in its protection. This byte concerns the access rights of the segment.

Virtual Address, Descriptor and Descriptor Table

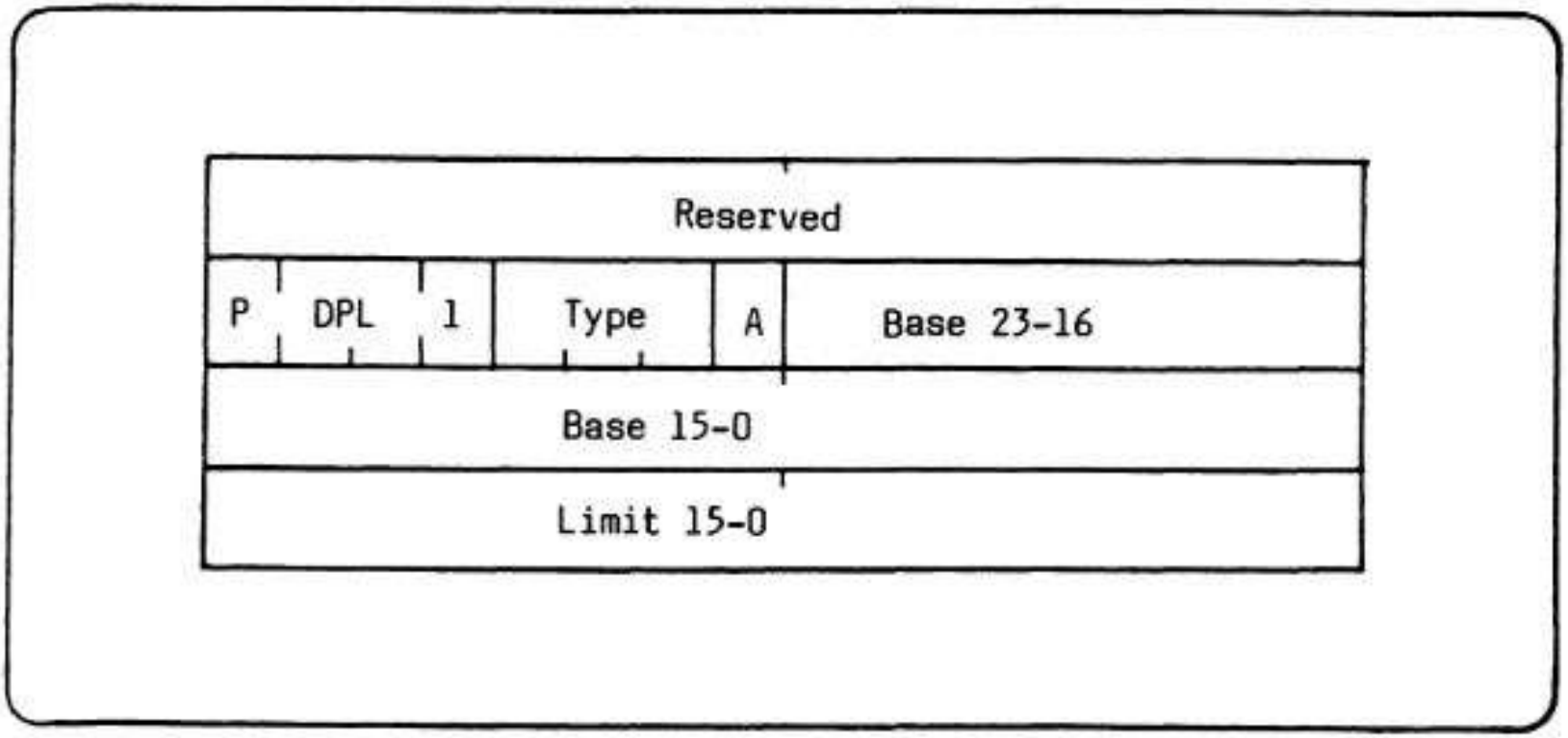
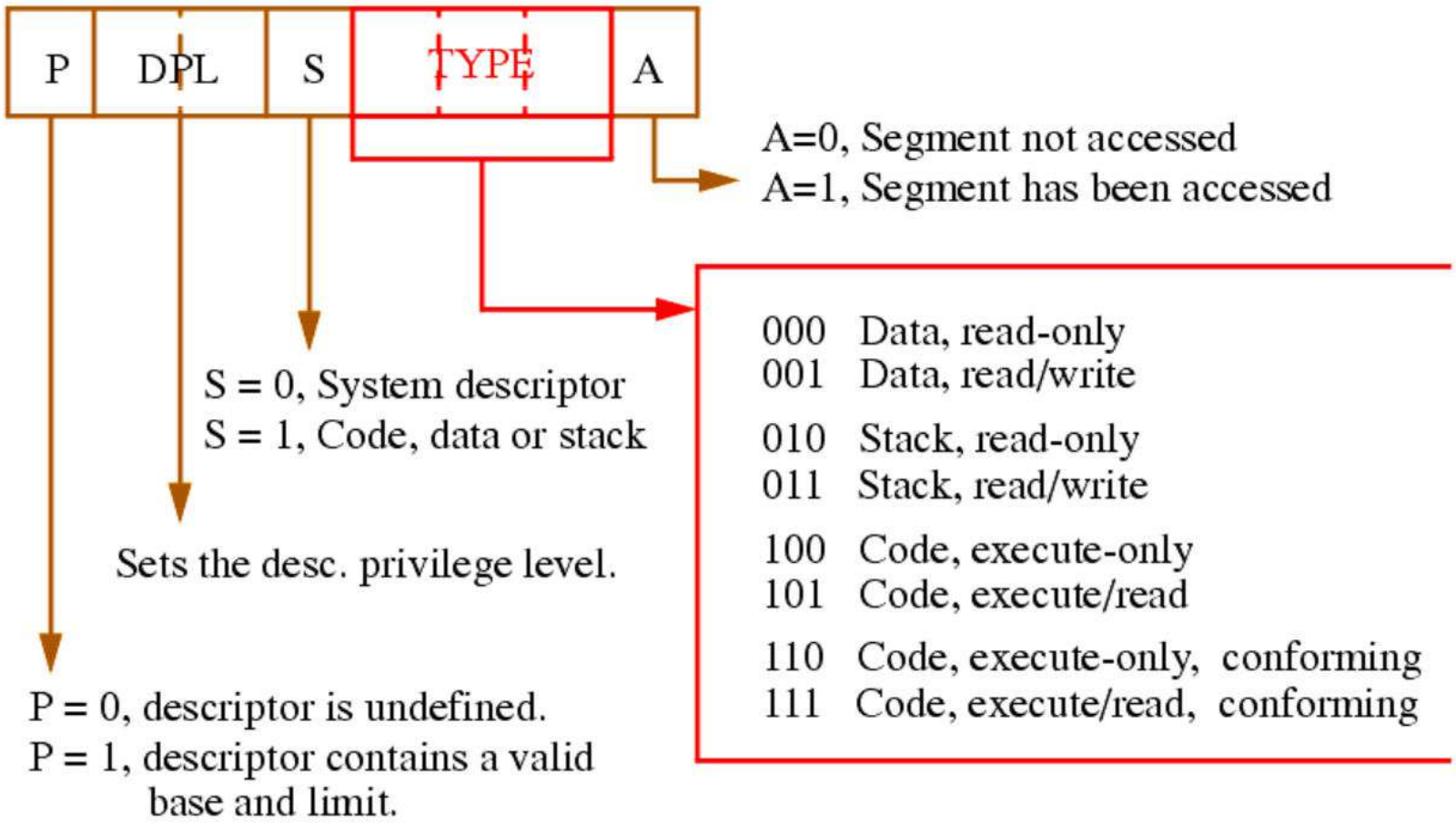


Figure 3.1.5 Segment descriptor

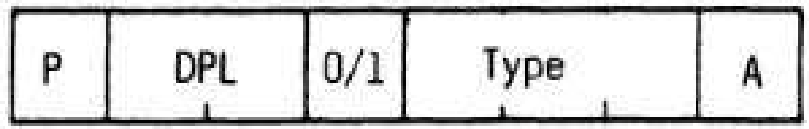
Virtual Address, Descriptor and Descriptor Table



Virtual Address, Descriptor and Descriptor Table

- The "P" bit defines the presence (1) or absence (0) of the segment from main memory.
  - ✓ This bit is used by the operating system process that migrates segments between main memory and secondary storage.
- The "DPL" bits define the rank or protection level attached to the segment, and is initialized by the Builder program during the final stage of program construction.
- The "A" bit is set by the hardware whenever an access is made to the segment.
- The full protection mechanisms of the 80286 are described in section 3.2. but it will be apparent from the above description of segment descriptors that at least two independent protection checks may be implemented preventing accesses to unallocated parts of segments, and preventing accesses to code and data at the wrong privilege levels.

Virtual Address, Descriptor and Descriptor Table



Segment type:
- read/write
- read only
- executable
- conformant
- 'expand down' (stack)

P indicates presence in main memory
A indicates segment has been accessed

Figure 3.1.7 Segment types

80386 Microprocessor

Features of 80386 processor

- As it is a 32-bit microprocessor. Thus has a 32-bit ALU.
- 80386 has a data bus of 32-bit.
- It holds an address bus of 32 bit.
- 80386 supports a variety of operating clock frequencies, which are 16 MHz, 20 MHz, 25 MHz, and 33 MHz.

Features of 80386 processor

- The 80386 microprocessor is an enhanced version of the 80286 microprocessor
- Memory-management unit is enhanced to provide memory paging.
- The 80386 also includes 32-bit extended registers and a 32-bit address and data bus. These extended registers include EAX, EBX, ECX, EDX, EBP, ESP, EDI, ESI, EIP and EFLAGS.
- The 80386 has a physical memory size of 4GBytes that can be addressed as a virtual memory with up to 64TBytes.
- The 80386 is operated in the pipelined mode, it sends the address of the next instruction or memory data to the memory system prior to completing the execution of the current instruction
- This allows the memory system to begin fetching the next instruction or data before the current is completed. This increases access time.

Features of 80386 processor

- The instruction set of the 80386 is enhanced to include instructions that address the 32-bit extended register set.
- The 80386 memory manager is similar to the 80286, except the physical addresses generated by the MMU are 32 bits wide instead of 24-bits.
- The concept of paging is introduced in 80386
- 80386 support three operating modes:
  1) Real Mode (default)
  2) Protected Virtual Address Mode (PVAM)
  3) Virtual Mode
- The memory management section of 80386 supports virtual memory, paging and four levels of protection.
- The 80386 includes special hardware for task switching.

Architecture of 80386 Microprocessor

- The internal architecture of the 80386 includes six functional units that operate in parallel. The parallel operation is called as pipeline processing.
- Fetching, decoding execution, memory management, and bus access for several instructions are performed simultaneously.
- The six functional units of the 80386 are
  1) Bus Interface Unit
  2) Code Pre-fetch Unit
  3) Instruction Decoder Unit
  4) Execution Unit
  5) Segmentation Unit
  6) Paging Unit

Architecture of 80386 Microprocessor

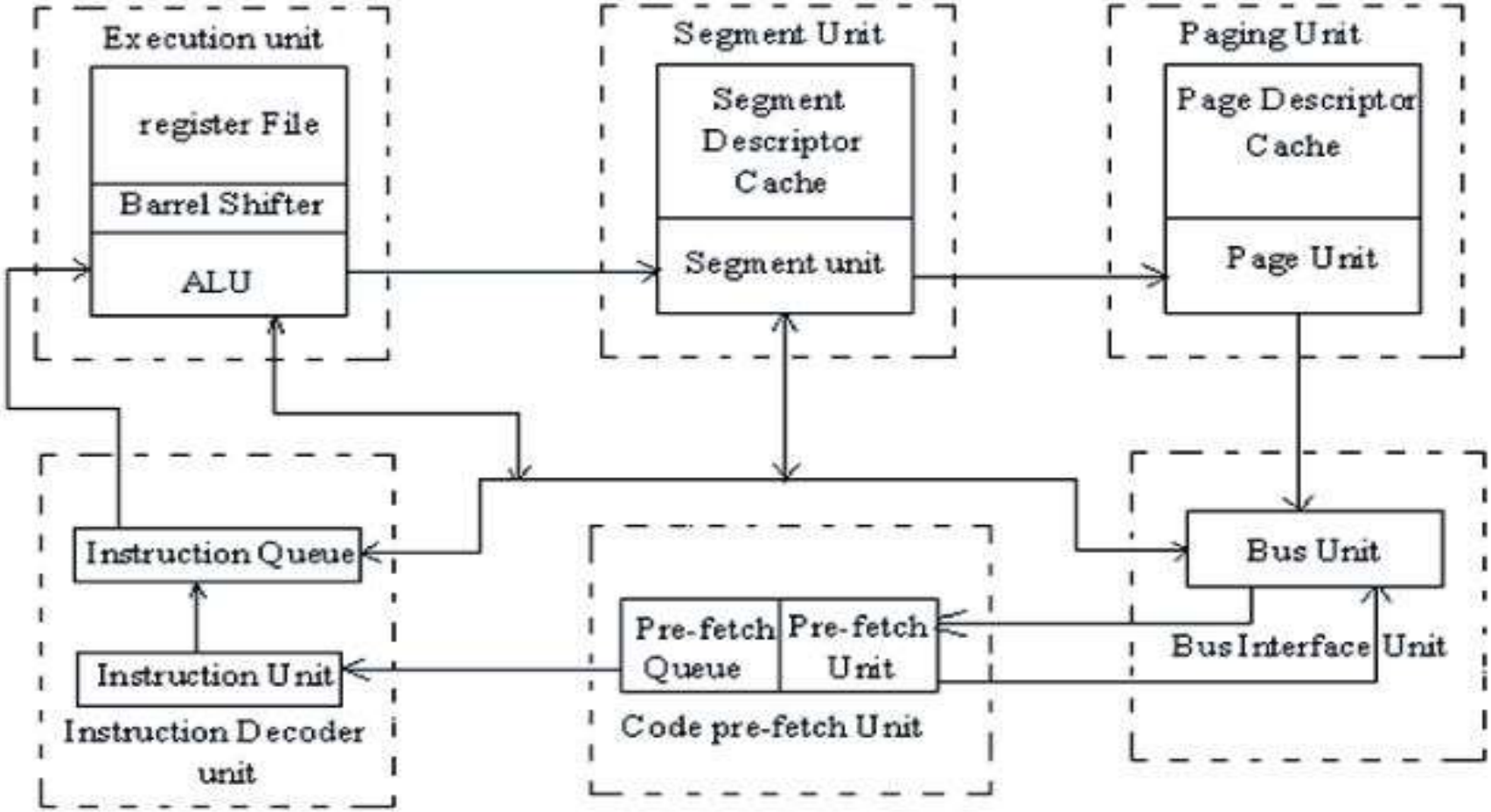


Figure: 80386 Architecture

Architecture of 80386 Microprocessor

- The Bus Interface Unit connects the 80386 with memory and I/O. Based on internal requests for fetching instructions and transferring data from the code pre-fetch unit, the 80386 generates the address, data and control signals for the current bus cycles.
- The code pre-fetch unit pre-fetches instructions when the bus interface unit is not executing the bus cycles. It then stores them in a 16-byte instruction queue for decoding by the instruction decode unit.
- The instruction decode unit translates instructions from the pre-fetch queue into micro-codes. The decoded instructions are then stored in an instruction queue (FIFO) for processing by the execution unit.
- The execution unit processes the instructions from the instruction queue. It contains a control unit, a data unit and a protection test unit.

Architecture of 80386 Microprocessor

- The control unit contains microcode and parallel hardware for fast multiply, divide and effective address calculation. The unit includes a 32-bit ALU, 8 general purpose registers and a 64-bit barrel shifter for performing multiple bit shifts in one clock. The data unit carries out data operations requested by the control unit.
- The protection test unit checks for segmentation violations under the control of microcode.
- The segmentation unit calculates and translates the logical address into linear addresses at the request of the execution unit.
- The translated linear address is sent to the paging unit. Upon enabling the paging mechanism, the 80386 translates these linear addresses into physical addresses.

- If paging is not enabled, the physical address is identical to the linear address and no translation is necessary.

Register Organization in 80386

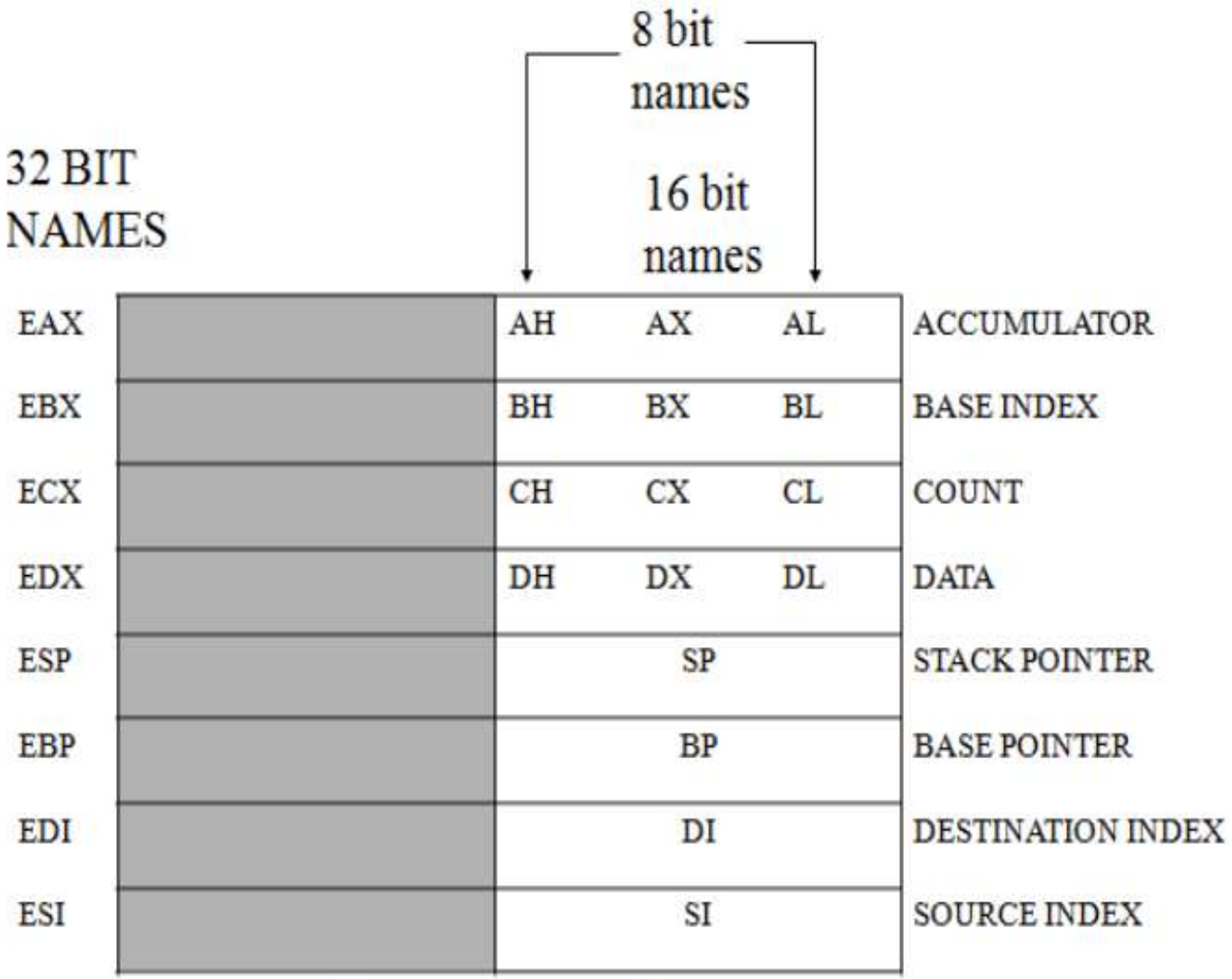


Figure:80386 General Purpose, Index and Pointer Register

General Purpose Registers

- Registers EAX, EBX, ECX, EDX, EBP, EDI and ESI are regarded as general purpose or multipurpose registers.
- **EAX (ACCUMULATOR)**:The accumulator is used for instructions such as multiplication, division and some of the adjustment instructions. In 80386 and above, the EAX register may also hold the offset address of a location in memory system.
- **EBX (BASE INDEX):** This can hold the offset address of a location in the memory system in all version of the microprocessor. It the 80386 and above EBX also can address memory data.
- **ECX (count):** This acts as a counter for various instructions.
- **EDX (data):** EDX is a general-purpose registers that holds a part of the result for multiplication or part of the division. In the 80386 and above this register can also address memory data.

Pointer and Index Registers

- **EBP (Base Pointer):** EBP points to a memory location in all version of the microprocessor for memory data transfers.

- **ESP (Stack Pointer):** ESP addresses an area of memory called the stack. The stack memory is a data LIFO data structure. The register is referred to as SP if used in 16 bit mode and ESP if referred to as a 32 bit register.
- **EDI (Destination index):** EDI often addresses string destination data for the string instruction. It also functions as either a 32-bit (EDI) or 16-bit (DI) general-purpose register.
- **ESI (Source index):** ESI can either be used as ESI or SI. It is often used to the address source string data for the string instructions. Like EDI ESI also functions as a general-purpose registers.

Segment Registers: CS, DS, SS, ES, FS and GS

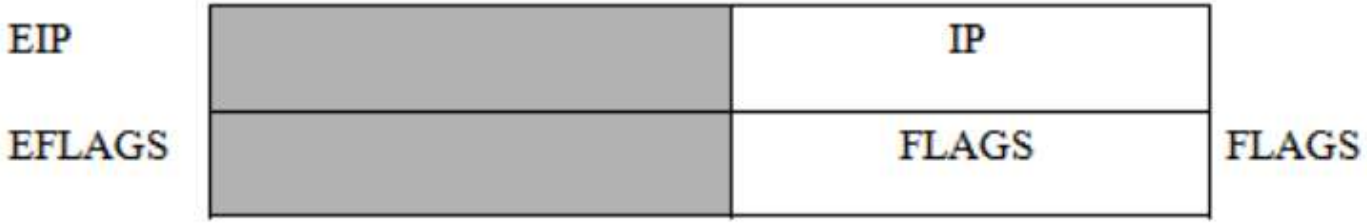
- **CS (Code):** The code segment is a section of memory that holds the code used by the microprocessor. The code segment registers defines the starting address of the section of memory holding code.
- **SS (Stack):** The stack segment defines the area of memory used for the stack. The stack entry point is determined by the stack segment and stack pointer registers. The BP registers also addresses data within the stack segment.
- **DS (Data):** The data section contains most data used by a program. Data are accessed in the data segment by an offset address of the contests of other registers that hold the offset address.

CS	CODE
DS	DATA
ES	EXTRA
SS	STACK
FS	
GS	

Figure:80386 Segment Register

Segment Registers: CS, DS, SS, ES, FS and GS

- **ES (extra):** The extra segment is used to hold information about string transfer and manipulation
- **FS and GS:** These are supplementing segment registers available in the 80386 and above microprocessors to allow two additional memory segments for access by programs.
- **EIP (Instruction Pointer):** EIP addresses the next instruction in a section of memory defined as a code segment. This register is IP (16bit) when microprocessor operates in the real mode and EIP (32 bits) when 80386 and above operate in protected mode



80386 Flag Register

Flag Register:

- Indicates the condition of the microprocessor and controls its operations. Flag registers are also upward compatible since the 8086-80268 have 16bit registers and the 80386 and above have EGLAF register (32 bits)

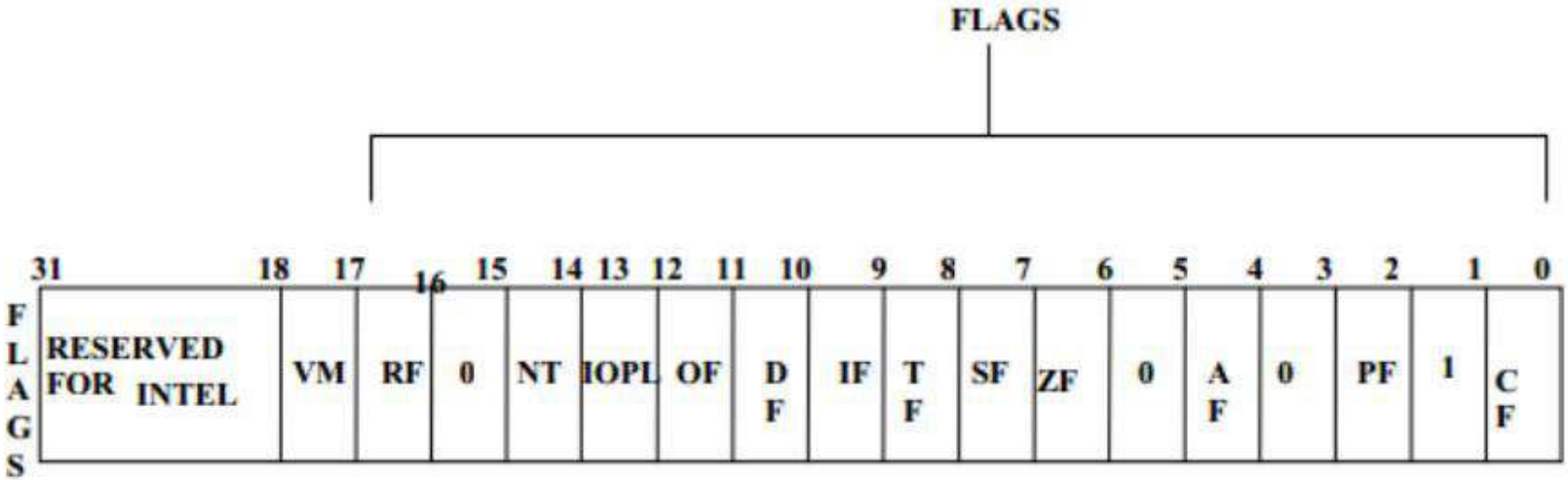


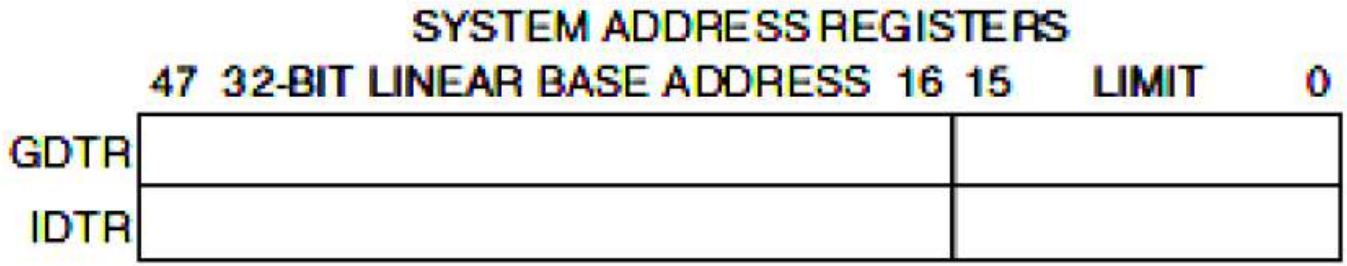
Figure: 80386 Flag Register

80386 Flag Register

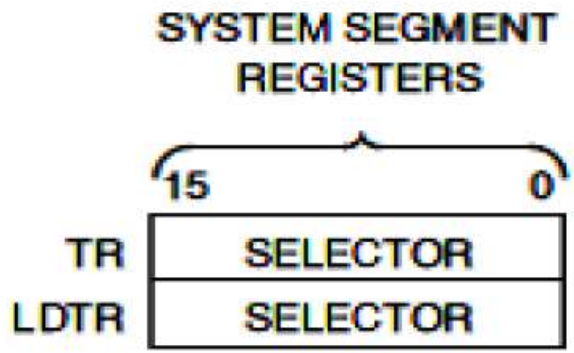
- **IOPL (I/O Privilege level):** IOPL is used in protected mode operation to select the privilege level for I./O devices. IF the current privilege level is higher or more trusted than the IOPL, I/O executed without hindrance. If the IOPL is lover than the current privilege level, an interrupt occurs, causing execution to suspend. Note that an IPOL is 00 is the highest or more trusted; if IOPL is 11, it's the lowest or least trusted.
- **NT (Nested Task):** The nested task flag is used to indicate that the current task is nested within another task in protected mode operation. This flag is when the task I nested by software.
- **RF (Resume):** The resume flag is used with debugging to control the resumption of execution after the next instruction.
- **VM (Virtual Mode):** The VM flag bit selects virtual mode operation in a protected mode system.

System Address Register

- Four memory management registers are used to specify the locations of data structures which control segmented memory management.



- GDTR (Global Descriptor Table Register) and IDTR (Interrupt Descriptor Table Register) be loaded with instructions which get a 6 byte data item from memory



- LDTR (Local Descriptor Table Register) and TR (Task Register) can be loaded with instructions which take a 16-bit segment selector as an operand.

80386 Special Register

Special 80386 Register

- Control Register: Four Control Register (CR0-CR3)
- Debug Register: Eight Debug Register (DR0-DR7)
- Test Register: Two Test Register (TR6-TR7)

Operating Modes of 80386

a. Real Address Mode
b. Protected Virtual Address mode
c. Virtual mode

Real address Mode

- Default Mode
- After reset, the 80386 starts from the memory location FFFFFFF0 H under real address mode.
- In real address mode, 80386 works as a fast 8086 with 32 bit registers and data types.
- Real-address mode is in effect after a signal on the RESET pin. Even if the system is going to be used in protected mode, the start-up program will execute in real-address mode temporarily while initializing for protected mode.
- The addressing techniques, memory size, interrupt handling in this mode of 80386 are similar to the real addressing mode of 80286.
- In real address mode, the default operand size is 16 bit but 32 bit operands and addressing modes may be used with the help of override prefixed.
- Maximum physical memory = 1Mega byte (1MB)
- The only way to leave real-address mode is to switch to protected mode.

Protected Virtual Address Mode

- Introduced in the 80386 processor.
- 32-bit address bus => access up to 232 bytes = 22. 230 B = 4 GB
- Base address => 32-bit value
- Offset =>16-bit or 32-bit value
- Linear address = base address + offset
- Linear address → physical address with paging
- In protected mode, the segment registers contain an index into a table of segment descriptors.
- Each segment descriptor contains the start address of the segment, to which the offset is added to generate the address.
- In addition, the segment descriptor contains memory protection information.
- This includes an offset limit and bits for write and read permission.
- This allows the processor to prevent memory accesses to certain data.
- Protected mode is accessed by placing a logic 1 into the PE bit of CR0
- This system contains one data segment descriptor and one code segment descriptor with each segment set to 4G bytes in length.
- PVAM mode support memory management, virtual memory, multitasking, protection, debugging, segmentation and paging.

Virtual Mode of 80386

- In its protected mode of operation, 80386DX provides a virtual 8086 operating environment to execute the 8086 programs.
- The real mode can also use to execute the 8086 programs along with the capabilities of 80386, like protection and a few additional instructions.
- Once the 80386 enters the protected mode from the real mode, it cannot return back to the real mode without a reset operation.
- Thus, the virtual 8086 mode of operation of 80386, offers an advantage of executing 8086 programs while in protected mode.
- The address forming mechanism in virtual 8086 mode is exactly identical with that of 8086 real mode.
- In virtual mode, 8086 can address 1Mbytes of physical memory that may be anywhere in the 4Gbytes address space of the protected mode of 80386.

80386 Paging Systems

- Paging is one of the memory management techniques
- The paging mechanism allows any linear (logical) address, as it is generated by a program, to be placed into any physical memory page, as generated by the paging mechanism.
- A linear memory page is a page that is addressed with a selector and an offset in either the real or protected mode.
- A physical memory page is a page that exists at some actual physical memory location.
- For example, linear memory location 20000H could be mapped into physical memory location 30000H, or any other location, with the paging unit. This means that an instruction that accesses location 20000H actually accesses location 30000H. Each 80386 memory page is 4K bytes long.
- Paging allows the system software to be placed at any physical address with the paging mechanism.
- Three components are used in page address translation: the page directory, the page table, and the actual physical memory page.

80386 Paging Systems

- The segmentation scheme may divide the physical memory into a variable size segments but the paging divides the memory into a fixed size pages.
- The segments are supposed to be the logical segments of the program.
- The pages are just fixed size portions of the program module or data.

80386 Paging Systems

Advantage:

- The advantage of paging scheme is that the complete segment of a task need not be in the physical memory at any time.
- Only a few pages of the segments, which are required currently for the execution need to be available in the physical memory.
- Thus the memory requirement of the task is substantially reduced, relinquishing the available memory for other tasks.

Page Directory

- The page directory contains the location of up to 1024 page translation tables, which are each four bytes long.
- Each page translation table translates a logical address into a physical address.
- The page directory is stored in the memory and accessed by the page descriptor address register (CR3).
- Control register CR3 holds the base address of the page directory, which starts at any 4K-byte boundary in the memory system.
- Each entry in the page directory translates the leftmost 10 bits of the memory address. This 10-bit portion of the linear address is used to locate different page tables for different page table entries.

Page Table

- The page table contains 1024 physical page addresses, accessed to translate a linear address into a physical address.

Page Directory Entry

- Total Page Directory Entries are1024
- Each directory entry is of 4byte

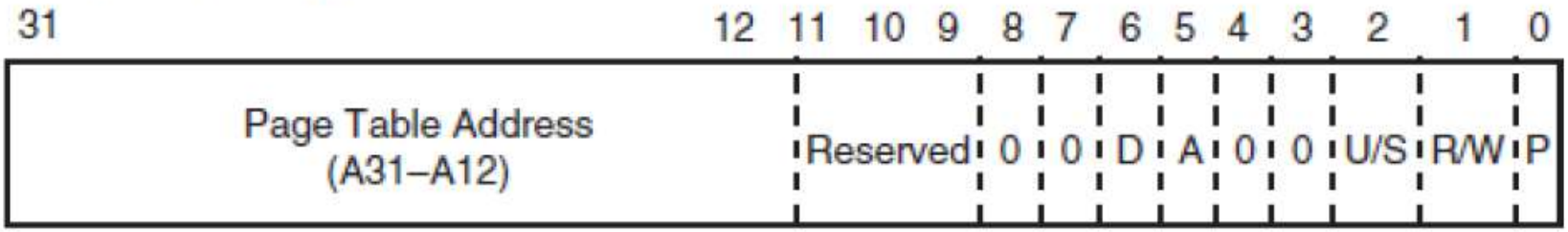


Figure: Page Directory Entry

Page Table Entry

- The page table entries contain the starting address of the page and the statistical information about the page.
- Total Entries are1024
- Each page table entry is of 4byte

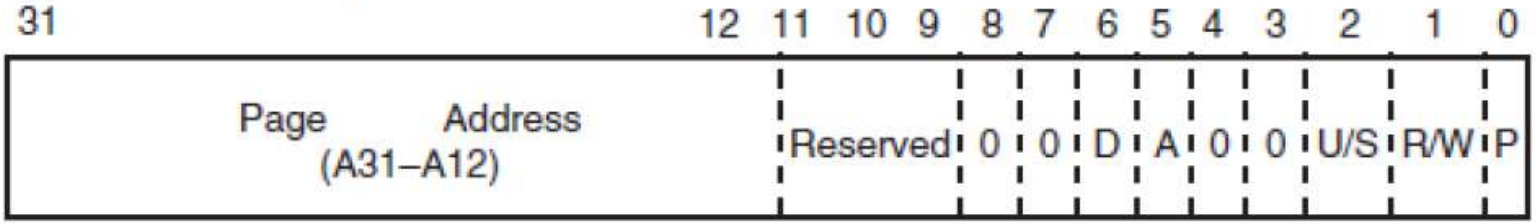


Figure: Page Table Entry

- **D-bit:** Dirty bit is undefined for page table directory entries by the 80386 microprocessor and is provided for use by the operating system.
- **A-bit:** Accessed bit is set to logic 1 whenever the microprocessor accesses the page Directory entry.

Page Table Entry

- **R/W** (Read/write) & **U/S** (user/supervisor) are both used in the protection scheme. Both bits combine to develop paging priority level protection for level 3, the lowest user level.

U/S	*R/W*	*Access Level 3*
0	0	None
0	1	None
1	0	Read-only
1	1	Write-only

- **P-bit:** Present bit, if logic1 indicates that the entry can be used in address translation. If P = 0, the entry cannot be used for translation. When P = 0, the remaining bits of the entry can be used to indicate the location of the page on the disk memory system.

Difference between Page directory and Page Table Entry

- The main difference is that the page directory entry contains the physical address of a page table, while the page table entry contains the physical address of a 4K-bytephysical page of memory.
- The other difference is the D (dirty bit), which has no function in thepage directory entry, but indicates that a page has been written to in a page table entry.

Page Translation Mechanism in 80386

- A page frame is a 4K-byte unit of contiguous addresses of physical memory. Pages begin on byte boundaries and are fixed in size.
- A linear address refers indirectly to a physical address by specifying a page table, a page within that table, and an offset within that page

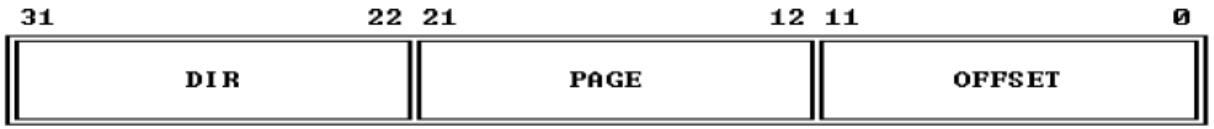


Figure: Format of Linear Address

- The below figure of page translation shows, how processor converts the DIR, PAGE, and OFFSET fields of a linear address into the physical address by consulting two levels of page tables.
- The addressing mechanism uses the DIR field as an index into a page directory, uses the PAGE field as an index into the page table determined by the page directory, and uses the OFFSET field to address a byte within the page determined by the page table.

Page Translation Mechanism in 80386

- In the second phase of address transformation, the 80386 transforms a linear address into a physical address.
- This phase of address transformation implements the basic features needed for page-oriented virtual-memory systems and page-level protection.
- Page translation is in effect only when the PG bit of CR0 is set.

Page Translation Mechanism in 80386

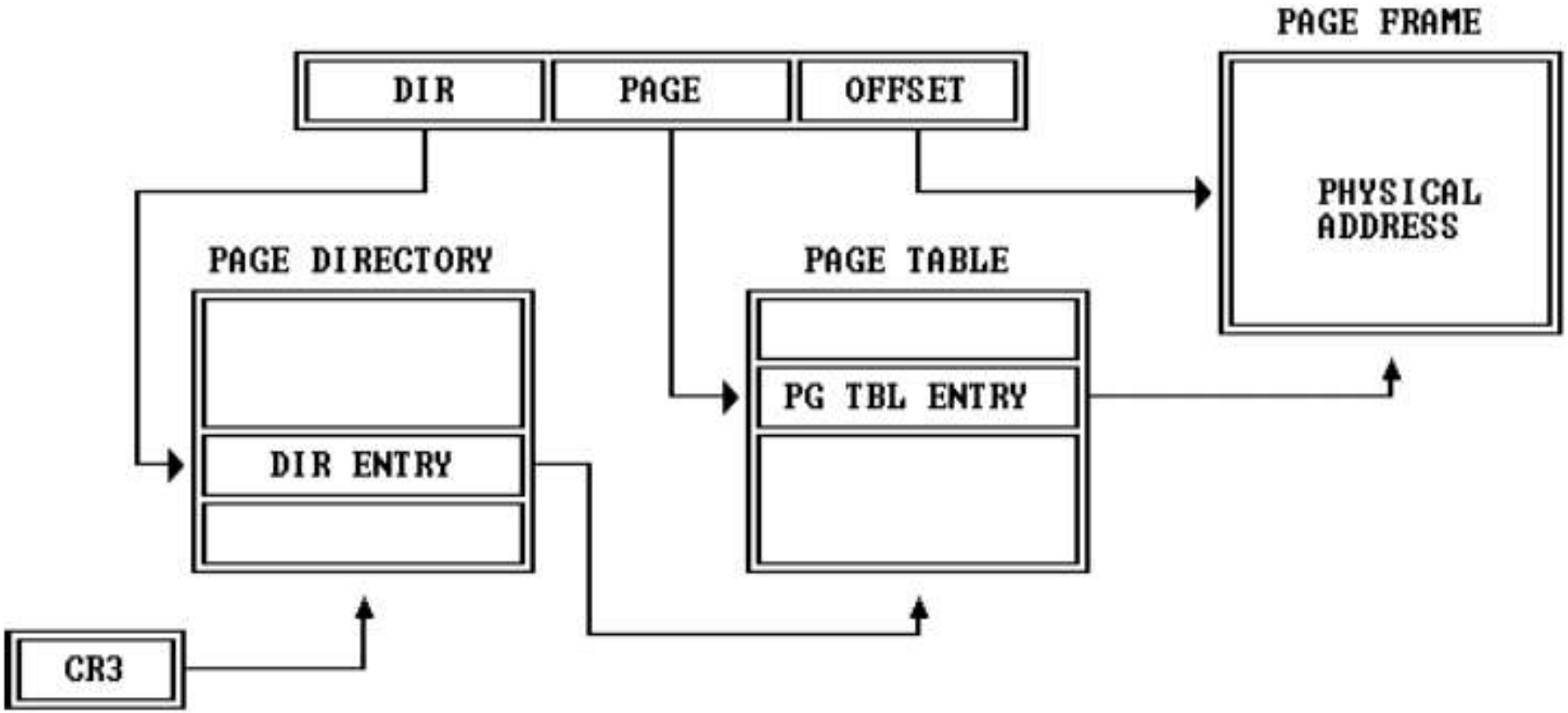


Figure: Page Translation